

ADITYA KAKDE

Software Architect | Senior Software Developer | Technical Product Lead – AI/ML Platforms

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PROFESSIONAL SUMMARY

Software architect and senior developer with 5+ years designing, building and shipping production AI/ML and GenAI platforms at ABB. I own features end to end, from specification and system design through backend services, APIs, UI and release, across microservices on Azure. Architected the Genix Copilot Foundation, a multilingual, multi-agent RAG platform for global industrial clients, and delivered platform-wide model explainability for Model Fabric, a microservices ML platform. De facto tech lead across two Scrum teams: I turn stakeholder needs into specs, roadmaps and shippable increments, run design and code reviews, and own production quality. Inventor on 6 patent-pending innovations in GenAI.

CORE SKILLS

Architecture & Engineering: System design & scalability · Microservices & API design (FastAPI) · Async/distributed processing (Celery) · Spec-driven design (SPEC, HLD/LLD) · Python · TypeScript/Angular · .NET (C#) · MongoDB · Parquet · Azure cloud · Helm · Ansible · CI/CD (Azure DevOps) · SonarQube · pytest

AI / ML & GenAI: LLM/RAG systems · Azure OpenAI · LangChain · LangGraph · Agentic AI · NL2SQL · Model explainability (SHAP) · ONNX inference · Vector databases · Azure Cognitive Search · LLMops/MLOps · MLflow · PromptFlow · Evaluation frameworks · AI-assisted developer tooling

Product & Leadership: Product roadmap & MVP definition · Requirements & spec writing · Stakeholder management & demos · Backlog prioritization · Agile/Scrum · Cross-functional leadership · Design & code reviews · Go-to-market alignment

EXPERIENCE

Software Engineer – GenAI, ML Platforms & LLMops

Jan 2024 – Present

ABB Ability Innovation Centre · Bengaluru, India

- **Technical leadership:** De facto tech lead across two Scrum teams. I make independent architectural decisions, lead design and code reviews, and own production stability and release quality from spec to deployment.

Model Fabric · *Microservices ML platform for inference, model registry, monitoring and optimization*

- **Model explainability (flagship feature):** Owned SHAP explainability end to end, from the spec package (SPEC, HLD/LLD, UI spec) to release. Built an async Celery SHAP worker in the ONNX Inference Engine, immutable Parquet persistence, a FastAPI /explain endpoint and Angular explanation panels localized into 7 languages, with no added inference latency.
- Extended model training to generate an explainability manifest (training stats, normal ranges), so users can see per-prediction feature contributions and out-of-range inputs directly in the dashboard.
- **Service decoupling:** Wrote a cross-module impact analysis and built a contract-compatible async task-result API that removed Model Registry's and Pipeline Designer's runtime dependency on the Model Recommender. Registry is now independently deployable, with zero functional regressions.
- **Quality engineering:** Restored CI quality visibility by fixing SonarQube coverage publishing for all modules and splitting a 2,900-line test monolith into ~10 focused modules. Wrote a full pytest contract suite for the FastAPI-migrated Optimization Engine (LP, PSO, FF optimizers).
- **Production reliability:** Eliminated silent data-lake prediction write-back failures with early config validation and clear error reporting. Stopped chart hangs on large datasets with single-pass filtering and 20k-point trace capping.
- **Customer-driven features:** Delivered configurable end-of-week resampling across settings, backend, UI and on-prem Ansible deployment, so customers can match aggregation to their business week. Built a demo app to align stakeholders on the change.
- **Innovation & enablement:** Built a .NET Edge AI proof of concept (model export and edge-device listing in the Model Registry UI). Set up AI-assisted developer tooling for the team: lean-code and debug agents, plus repo-map and commit/PR skills.

Genix Copilot Foundation & Genix APM · *Multilingual, multi-agent GenAI copilot platform*

- Architected and led end-to-end delivery of Genix Copilot Foundation, a production multilingual RAG platform on Azure OpenAI serving industrial IoT clients globally, owning system design, deployment and production quality.
- Designed a LangGraph-based agentic framework with dependency injection, cutting agent and tool onboarding time for consuming teams and enabling custom agentic flows without breaking existing integrations.
- Engineered an NL2SQL module with intent-based smart caching, cutting repeat-query latency by ~75% and improving natural-language precision by 60%.
- Built a PromptFlow-based LLM evaluation framework (BLEU, semantic similarity), creating a continuous quality pipeline across all GenAI use cases.

- Engineered a large-scale document indexing pipeline with delta-aware refresh, 5-snapshot index versioning and cross-environment promotion, cutting indexing time by ~40% and enabling zero-downtime rollbacks.
- Developed an Event Rationalization module for Genix APM Predict 360 on Azure OpenAI that computes risk scores and applies hierarchical alert suppression, reducing critical alert fatigue for plant operators.

Data Science Engineer

Sep 2020 – Sep 2022

ABB Ability Innovation Centre · Bengaluru, India

- Reduced manufacturing downtime by 34% with a deep learning-based digital twin that detects process anomalies on a stainless-steel production line, deployed in a live industrial environment.
- Built division-level Power BI dashboards for global orders, revenue and market analytics, informing marketing strategy and contributing \$12K in annual business impact through better customer-service targeting.
- Delivered the full data science lifecycle on both projects: data pipeline design, model development, validation and stakeholder reporting.

PATENTS & INNOVATIONS (6 PENDING · INVENTOR)

- **AI Copilot for Energy & Carbon Tracking** — Natural-language querying of real-time meter and production data for KPI generation, anomaly detection and carbon simulation in ESG reporting.
- **Smart Caching for NL2SQL Copilot Optimisation** — Detects intent similarity and reuses cached SQL to cut LLM cost and latency without losing accuracy.
- **Large-Scale Document Indexing with Dynamic Chunking** — Adaptive chunking and embeddings for indexing large corpora into vector databases, improving retrieval quality and latency.
- **Redis-Based Similarity Search for NL2SQL Mapping** — Cosine similarity over Redis-stored question-query pairs to reuse the closest SQL and reduce LLM round trips.
- **Evaluation Framework for Generative AI with PromptFlow** — BLEU and semantic-similarity validation of LLM outputs against ground truth at scale.
- **Multilingual Translation Copilot for Global Language Support** — Translation workflows and context-preserving embeddings that extend copilots to global users without retraining.

EDUCATION

Master of Science in Data Science

2023

Northeastern University · Boston, USA

Bachelor of Engineering in Computer Science

2020

Visvesvaraya Technological University · Belagavi, India